Supplementary Material: Directional Prompting for Gaze-Referential Inference in Large Multimodal Models

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This document accompanies the MMM 2027 submission. It collects dataset-construction details, prompt templates, additional tables, qualitative examples, and implementation hyperparameters. The main paper is self-contained; this file is for reproducibility.

1 Dataset Construction

FG-GRI. FG-GRI is derived from the 907-image public Stimuli release of Zhang et al. [6]. We retain 904 images and exclude Stimulus_ID 229, 230, and 867 because the observer's eyes are not visible. Each retained instance uses the original RGB image, candidate list, and ground-truth gazed object. Its ground-truth arrow uses a human-annotated start at the observer's eyes and a human-annotated endpoint at the gazed object.

CG-GRI (N=483) is derived from VideoAttentionTarget [1] (N=200) and GazeFollow [3] (N=283). Neither source provides GRI-ready candidate lists, so we construct MCQ options and targets as follows. We write *VAT-GRI* and *GF-GRI* for these two subsets of CG-GRI; they are reporting slices of a single track, not additional evaluation tracks.

VideoAttentionTarget (VAT-GRI). VideoAttentionTarget consists of video clips with dense per-frame gaze-target annotations. Consecutive frames within a clip are visually similar, so we sample by gaze sequence / clip and keep at most one frame per clip for the frozen experimental release. Automatic screening uses YOLOv10L [4] detections: the observer head box must be valid; detections overlapping the observer box are removed; the annotated gaze point must fall in at least one remaining box; and at least two candidate boxes remain. Short natural-language descriptions of the observer and each candidate are then written (LMM draft + human edit) so that instance phrases are visually distinguishable (e.g. two cups described by appearance or relative position, not by box coordinates). Every retained row is human-reviewed: observer identity, candidate list, and the target implied by the original VAT gaze point. Ambiguous or unverifiable images are discarded. Ground-truth arrow starts and endpoints are human-annotated: each start is a reviewed eye point (accepted or clicked), and each endpoint is the original VAT gaze point. The ground-truth direction is therefore the ray from the annotated eye point to the annotated gaze point. This yields the frozen **200** VAT-GRI instances used in the main paper.

GazeFollow (GF-GRI). GazeFollow provides web images annotated with gazed-region heatmaps but not with gazed-object labels. From an initial sample of 1,000 images, we retain, by manual judgment, only those in which the observer's eyes are clearly visible. We discard images in which the observer looks away from the camera, the gaze direction is occluded, or eye visibility is unclear. For each retained image we (1) list candidate objects (YOLO draft + human merge/add) and (2) label the ground-truth gazed target from the heatmap. Ground-truth arrows use the original GazeFollow eye annotation as the origin and the original gaze point as the endpoint. This yields **283** GF-GRI instances.

Shared protocol. On CG-GRI the observer is marked with an unfilled red bounding box to disambiguate multi-person scenes; the box is *not* used as the arrow origin. It serves two other purposes. It is drawn in every CG-GRI condition, including the RGB baseline, so that the only difference between conditions is the directional cue. It also selects whose gaze to predict: among the faces detected in the full image, we take the largest one overlapping the box as a reference head, run the gaze estimator once per overlapping face with a tight box around it, and keep the run whose projected origin lands closest to that reference head. The origin itself always comes from the estimator, never from the box. FG-GRI does not use a head box: it uses the original candidate labels, and its ground-truth arrow uses the human-annotated eye start and gaze endpoint. The multiple-choice question lists the verified candidate names and asks which object the observer is looking at. Ground-truth and UniGaze arrows (and matched textual coordinates) are built from the same arrow record.

2 Prompt Templates

All GRI conditions share the same zero-shot MCQ stem and JSON parser unless the condition *is* a prompt change (textual coordinates, CoT, CoGT, or LMM-generated Stage 1). Answers are parsed from the JSON `answer` field and scored by case-insensitive exact string match against the ground-truth label; a missing, malformed, or non-candidate answer is scored incorrect rather than imputed. `{choices}` is replaced at runtime with candidate names, one per line. Candidate order is fixed for an instance across all conditions.

FG-GRI (plain). Used for RGB, D, RGB+D, RGB+A, D+A, RGB+D+A, and Stage 2 of the LMM-generated-arrow protocol (the overlay already contains the arrow).

Prompt (FG-GRI)

```
What is the person looking at in the image? You must choose
exactly one option from the following options:
{choices}
Please answer in the following format:
```json{"answer": "<your chosen option>", "reasoning": "<your
reasoning>"}```
```

CG-GRI (plain). Same stem, but the observer is identified by the red bounding box.

Prompt (CG-GRI)

```
What is the person in the red bounding box looking at?
You must choose one from the following options:
{choices}
Please answer in the following format:
```json{"answer": "<your chosen option>", "reasoning": "<your
reasoning>"}```
```

*Textual directional prompting (*RGB+A_{txt}*).* The image remains RGB, with the observer box retained on CG-GRI. The same start–end geometry as the gaze-direction overlay is inserted after the question and before the choice list. Ground-truth coordinates are used in the GT variant; UniGaze-predicted coordinates are used in the UniGaze variant. On CG-GRI the first sentence uses "the person in the red bounding box" as above.

Prompt with textual arrow (RGB+A_{txt}, FG-GRI wording)

```
What is the person looking at in the image? The target person's
gaze ray originates from pixel ({x1}, {y1}) and ends at pixel ({x2},
{y2}). The origin (0, 0) is at the top-left corner of the image.
You must choose exactly one option from the following options:
{choices}
Please answer in the following format:
```json{"answer": "<your chosen option>", "reasoning": "<your
reasoning>"}```
```

CoT and CoGT (FG-GRI). Both conditions retain the plain FG-GRI MCQ stem and JSON answer instruction above. CoT appends "Let's think step by step." CoGT instead appends "Let's think step by step. Pay attention to the target person's eyes, imagine the gaze direction as an arrow line, and follow the line to determine the gazed object." As in the main paper, CoGT uses RGB only and is not combined with RGB+A.

LMM-generated arrow, Stage 1. Stage 1 sees RGB only (plus the observer box on CG-GRI), *without* answer choices, and must emit a gaze arrow. A deterministic
```

renderer overlays that arrow; Stage 2 then uses the plain MCQ prompt above. Malformed rays and API failures are recorded as failures and are never replaced by a UniGaze or ground-truth arrow. FG-GRI validates the returned coordinates but applies no length floor; all three models produce a valid arrow for all 904 images. On CG-GRI, denormalized rays shorter than 35 px are also failures. Gemini-3.1-pro, GPT-5.2, and Qwen3.5-27B produce 1, 8, and 15 such rays, respectively; Qwen has two additional Stage 1 API failures. All Stage 1 failures are scored incorrect and receive no Stage 2 answer.

The RGB+LMM-gen accuracy is computed over all 483 CG-GRI instances for every model. Paired comparisons with RGB+A(UniGaze) use N=483 for Gemini and GPT-5.2, and N=481 for Qwen because two UniGaze overlays are missing; a remaining Stage 1 API failure stays in that paired set and is scored incorrect.

LMM-generated Stage 1 (FG-GRI)

```
Identify the target person's gaze direction in this image. Do not
name an object.
Return exactly one JSON object, using a normalized 0-1000 image
coordinate system: top-left is (0,0), bottom-right is (1000,1000).
The ray start must be near the target person's visible eyes. The ray
end must be where that person is looking.
Format: {"start": [x, y], "end": [x, y]}
All four values must be numbers between 0 and 1000 inclusive.
```

LMM-generated Stage 1 (CG-GRI)

```
The person whose gaze you must estimate is marked by a red bounding
box.
Infer that person's gaze direction from the image. Do not identify an
object and do not list any answer choices.
Return exactly one JSON object and nothing else. Use a coordinate
system in which the top-left is (0, 0) and the bottom-right is (1000,
1000):
{"start": [x, y], "end": [x, y]}
Both points must be within 0 through 1000. The start must be near the
observer's eyes and the end must lie in the gaze direction.
```

Point / line controls. The endpoint-only condition renders a red point at the predicted arrow endpoint. The headless-line condition renders the same red shaft with the same origin and endpoint as RGB+A, but without an arrowhead. Both use the same plain MCQ prompt as RGB+A; only the overlay rendering changes.

3 Predicted-Arrow Construction and Endpoint Rule

From a 3D gaze direction to a 2D arrow. UniGaze returns a 3D gaze vector, which we project into the image before any depth reasoning. Six facial landmarks—the four eye corners and the two nose corners—are fitted to a 3D

face model by `solvePnP` under a fixed pinhole camera (focal length $4W$ for image width W, principal point at the image center, no distortion), which gives the head rotation and translation. The arrow origin is the projection of the resulting 3D face center, the mean of those six landmarks, which is also the point UniGaze regresses its direction from. Projecting a second 3D point at a fixed distance along the predicted vector gives a hint pixel that fixes only the 2D *direction* of the arrow; its length comes from the depth map, so the choice of that distance does not affect the rendered cue.

Terminating the ray on the depth map. The 2D ray is extended from the origin through the hint pixel to the image border, whose intersection (u_e, v_e) serves as a reference. Writing z_0 and z_e for the Depth Anything V2 values at the origin and at (u_e, v_e), we sample the ray every 0.75 px between 10 px (skipping the observer's own face) and 800 px, and compare the scene depth $z_{\text{scene}}(t)$ with a ray depth interpolated linearly in image space, $z_{\text{ray}}(t) = z_0 + \frac{t}{\|(u_e,v_e)-(u_0,v_0)\|}(z_e - z_0)$. Depth Anything V2 predicts relative inverse depth, so larger values are nearer and the ray is blocked wherever $z_{\text{ray}}(t) \leq z_{\text{scene}}(t) + \tau$, with the tolerance τ set to 2% of the image's valid depth range (1st–99th percentile). The first samples of the maximal contiguous intervals satisfying this test are the first and second occluders; the border intersection is used when no interval exists. The test is deliberately relative: it needs neither metric depth nor camera calibration beyond the fixed pinhole model above.

Ground-truth arrows. Ground-truth visual and textual arrows are prepared for every retained instance on both tracks. On FG-GRI both the start and the endpoint are human-annotated. On CG-GRI the source-specific origins and endpoints are described in Section 1.

Predicted arrows. Predicted arrows take their start from the projected UniGaze face center and their endpoints from UniGaze direction plus first-occluder depth grounding. On FG-GRI, the GT arrow is therefore the human-annotated eye-to-target ray, whereas the UniGaze arrow is a predicted ray; they are not constrained to share an origin. For D+A, endpoint depth semantics also differ: the GT variant uses scene depth at the annotated endpoint, whereas the UniGaze variant uses the interpolated ray depth at the occluder. FG-GRI is built from the pilot stimuli of Zhang et al. [6]; their public release contains 907 images, whereas the paper describes the pilot pool as 900.

Which occluder to use. A UniGaze-predicted arrow is terminated on the monocular depth map at the *first occluder* (physical default: first intersection with estimated scene geometry), the *second occluder*, or the *image boundary*. The comparison is a sensitivity analysis of that default, not a test-set choice of method. On FG-GRI (N=904), Gemini-3.1-pro and Qwen3.5-27B rank the three rules the same: first occluder (86.4% / 72.2%) ahead of the second occluder (82.2% / 69.0%) and the image boundary (80.1% / 62.4%). For Gemini the pairwise

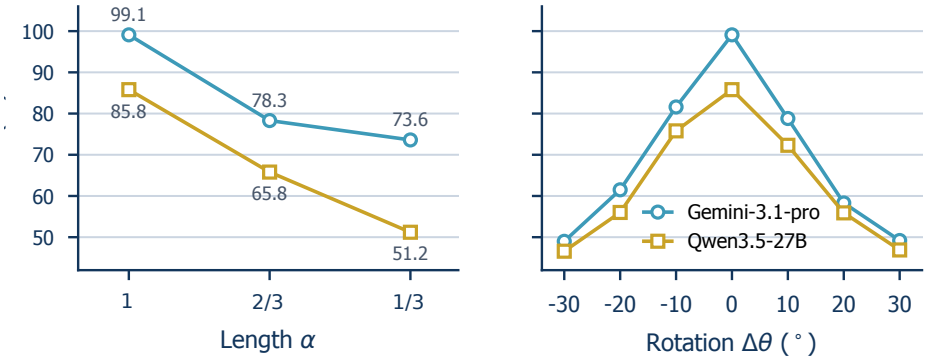


Fig. 1. Sensitivity of FG-GRI accuracy to ground-truth arrow length and direction for `Gemini-3.1-pro` and `Qwen3.5-27B` under RGB+A(GT).

drops are Δ=−4.2 pp (p=0.0018) and Δ=−6.3 pp ($p<10^{-5}$); both models' pairwise differences from the default are significant (paired McNemar, $p<0.05$). Even boundary termination remains well above each model's RGB baseline (55.5% / 44.6%). One reason first-occluder ranks highest is specific to FG-GRI: tabletop scenes place the labeled target on the nearest surface along the ray, so the first occluder usually coincides with the annotated endpoint. Reported UniGaze RGB+A numbers keep this physically motivated default; we do not treat it as universally optimal.

4 Arrow Length and Direction Sensitivity

We further test whether FG-GRI accuracy is sensitive to errors in the *geometry* of a ground-truth visual arrow, isolating cue quality from UniGaze prediction noise. On Gemini-3.1-pro and Qwen3.5-27B, we shorten the arrow to α of its original length with $\alpha \in \{1, \frac{2}{3}, \frac{1}{3}\}$, or rotate it by $\Delta\theta \in \{-30, -20, -10, 0, 10, 20, 30\}$ degrees about the arrow origin while keeping the target label unchanged. Figure 1 shows that accuracy falls monotonically as the arrow shortens and as rotation increases in magnitude; every perturbed condition differs significantly from the unperturbed arrow (paired McNemar, all $p < 0.01$). The same pattern appears for both evaluated LMMs.

5 LMM-Generated Arrow: Gaze Alignment on CG-GRI

Table 1 reproduces the main paper's CG-GRI comparison between UniGaze and LMM-generated arrows. The sign differs by model: Gemini-3.1-pro reaches 81.8%, above its 77.2% UniGaze accuracy (+4.6 pp over paired instances; McNemar p=0.068), whereas Qwen3.5-27B and GPT-5.2 fall below their UniGaze

Table 1. CG-GRI accuracy under UniGaze and LMM-generated arrows (main-paper spatial-cue table, CG-GRI rows). All RGB+LMM-gen accuracies use N=483. Qwen3.5-27B's UniGaze accuracy, difference, and paired test use N=481; the other paired comparisons use N=483.

Model	RGB+A(UniGaze)	RGB+LMM-gen	Δ	McNemar p
Qwen3.5-27B	77.5	72.0	−5.5	0.012
GPT-5.2	76.2	72.0	−4.1	0.054
Gemini-3.1-pro	77.2	81.8	+4.6	0.068

Table 2. Median gaze alignment of LMM-generated and UniGaze arrows. Each UniGaze row uses the same instances as the corresponding LMM-generated row; accuracies are the full-CG totals from Table 1.

Cue / model	Angle (°)	Endpoint dist. (px)	Acc. (%)
Gemini LMM-gen	4.8	36	81.8
Gemini UniGaze	12.3	152	77.2
GPT-5.2 LMM-gen	10.4	188	72.0
GPT-5.2 UniGaze	12.1	154	76.2
Qwen3.5-27B LMM-gen	9.1	98	72.0
Qwen3.5-27B UniGaze	12.3	155	77.5

results. This section documents a post-hoc audit that asks whether those CG-GRI differences co-vary with how well each overlay aligns with the ground-truth gaze ray. It is exploratory and does not change any main-paper numbers.

Metrics. For each instance we compute (i) *angular error*, the angle between the overlay ray and the ground-truth gaze direction, and (ii) *endpoint distance*, the Euclidean distance from the overlay endpoint to the annotated ground-truth gaze point (pixels). Medians use instances for which both the LMM-generated ray and the paired UniGaze ray can be compared to the annotated gaze point. In Table 1, all RGB+LMM-gen accuracies use the full 483 instances; Qwen's UniGaze accuracy and paired test use the 481-instance overlap described above.

Coordinate conversion. LMM-generated Stage-1 rays are emitted on a normalized 0–1000 grid (Section 2). Before any distance or angle computation, self-generated endpoints are mapped to pixel space as $x_{\text{px}} = x_{\text{norm}}(W-1)/1000$ (and analogously for y), matching the released renderer. GazeFollow ground-truth and UniGaze coordinates are stored in $[0, 1]$ and scaled by image width and height; VideoAttentionTarget gaze points are already in pixels. All comparisons use the ground-truth *gaze point*, not candidate bounding boxes.

Pooled alignment. Table 2 shows that Gemini self-generated endpoints are closer in median distance to the ground-truth gaze point than UniGaze overlays on the same subset (36 px vs. 152 px) and carry smaller median angular error (4.8° vs. 12.3°). Qwen3.5-27B also has lower median endpoint distance and angular error with its self-generated rays, but this geometry advantage does not translate

Table 3. Gemini-3.1-pro median gaze alignment by CG-GRI source.

Source	UniGaze dist. (px)	Self dist. (px)	UniGaze angle (°)	Self angle (°)
VAT-GRI	286	46	11.4	5.4
GF-GRI	95	28	12.9	4.2

Table 4. Gemini-3.1-pro: median gaze alignment by RGB+A(UniGaze)→RGB+LMM-gen transition type. Acc. is self (LMM-gen) accuracy within each row.

Transition	Self angle (°)	UniGaze angle (°)	Self dist. (px)	UniGaze dist. (px)	Self acc. (%)
fix	5.4	15.3	28	174	100
break	66.3	8.4	336	192	0
stable_correct	3.5	12.2	26	131	100
stable_wrong	18.3	9.3	391	237	0

into an accuracy gain. GPT-5.2 self-generated endpoints are farther from the gaze point than its paired UniGaze endpoints (188 px vs. 154 px). Table 3 splits Gemini by source: on both VAT-GRI and GF-GRI, self-generated endpoints are closer to the annotated gaze point than UniGaze (VAT: 46 vs. 286 px; GF: 28 vs. 95 px).

Instance transitions. To relate accuracy changes to cue quality, we classify each paired RGB+A(UniGaze)→RGB+LMM-gen instance by whether UniGaze and the self-generated overlay agree: *fix* (UniGaze wrong, self correct), *break* (UniGaze correct, self wrong), *stable_ correct*, or *stable_ wrong*. On full CG-GRI (N=483), Gemini-3.1-pro has 77 fixes and 55 breaks under this definition, consistent with its net +4.6 pp gain. Table 4 reports Gemini's median alignment by transition type. *Fix* cases combine better self alignment (median 5.4° angular error, 28 px endpoint distance) with UniGaze errors (15.3°, 174 px). *Break* cases reverse the pattern: UniGaze remains well aligned (8.4°, 192 px) while self-generated direction error explodes (median 66.3°), yielding 0% self accuracy by construction. Gemini's CG-GRI gain is therefore fix-heavy and consistent with better self-drawn gaze geometry on instances where UniGaze fails, not with uniformly better pooled endpoint placement alone.

Angular-error bins (exploratory). Gemini self accuracy rises to 95.7% when self angular error is $\leq$ 10° but falls to 49.4% when error exceeds 30°, whereas UniGaze accuracy varies much less across the same bins (82.3% vs. 73.6%).

6 Error-Type Examples and Stratified Distribution

Figure 2 shows one representative FG-GRI failure for each of the four error types under RGB+A(UniGaze) (Gemini-3.1-pro). (a) *Inaccurate cue*: the UniGaze arrow has the wrong direction and/or endpoint. (b) *Geometric ambiguity*: several candidates lie near the tip. (c) *Reasoning hallucination*: the LMM's rationale

contradicts the overlay. (d) *Depth-order confusion*: the arrow already points at the labeled object, but a nearer-to-camera or more salient candidate is chosen. The main paper reports the aggregate counts; each of the 123 failures was labeled by one author using the four definitions above. Table 5 stratifies the same failures by view, proximity, and number of candidates. Inaccurate cues dominate almost every bucket; geometric ambiguity, reasoning hallucination, and depth-order confusion remain rare. Right-view errors show a higher share of reasoning hallucination (18.5%) than left-view errors (3.0%).

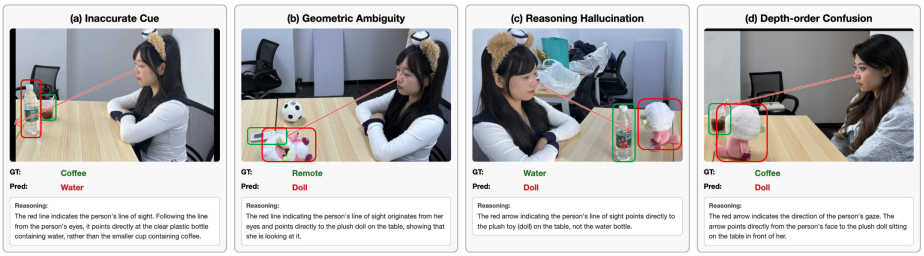


Fig. 2. Representative FG-GRI errors for Gemini-3.1-pro with RGB+A(UniGaze). Green and red boxes denote the ground-truth and predicted answers, respectively.

Table 5. Stratified error-type distribution (`Gemini-3.1-pro`, RGB+A(UniGaze), FG-GRI; N=123 failures). Each cell: count (%).

Stratification	(a) Inaccurate Cue	(b) Geometric Ambiguity	(c) Reasoning Hallucination	(d) Depth-order Confusion
By View Angle				
Left (N=66)	59 (89.4%)	2 (3.0%)	2 (3.0%)	3 (4.5%)
Right (N=54)	37 (68.5%)	4 (7.4%)	10 (18.5%)	3 (5.6%)
Front (N=3)	3 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
By Proximity				
Level 1 (closest, N=68)	52 (76.5%)	6 (8.8%)	8 (11.8%)	2 (2.9%)
Level 2 (medium, N=30)	24 (80.0%)	0 (0.0%)	4 (13.3%)	2 (6.7%)
Level 3 (farthest, N=25)	23 (92.0%)	0 (0.0%)	0 (0.0%)	2 (8.0%)
By Number of Options				
2 options (N=60)	45 (75.0%)	5 (8.3%)	7 (11.7%)	3 (5.0%)
3 options (N=34)	26 (76.5%)	1 (2.9%)	4 (11.8%)	3 (8.8%)
4 options (N=29)	28 (96.6%)	0 (0.0%)	1 (3.4%)	0 (0.0%)

7 Limitations of Depth as an LMM Input

Adding a depth-color arrow on a depth plate (D+A) yields limited and inconsistent gains relative to adding a gaze-direction arrow on RGB (main-paper overall table). On FG-GRI, Gemma-4-26B improves only +3.8 pp from D to D+A(GT) (39.9%→43.7%), far below +41.7 pp from RGB to RGB+A(GT) (50.8%→92.5%). Candidate objects and the pseudo-color shaft are harder to recognize on depth plates (main-paper Figure 1); some LMMs describe depth images as thermal or infrared scenes.

An occasional D+A(UniGaze)>D+A(GT) gap on depth plates should not be read as UniGaze outperforming manual gaze in general: on RGB+A, GT arrows dominate for every model. On FG-GRI D+A, UniGaze arrows exceed GT-arrow accuracy for six of nine LMMs (largest for Gemma-4-26B, Δ=+7.2 pp, McNemar $p<0.001$), which we attribute to colormap and first-occluder artifacts in brittle depth rationales rather than a better 3D gaze estimate.

8 Gaze-Informed Reasoning Failure Examples

We inspect Gemini-3.1-pro failures on FG-GRI when a structured prompt inspired by GazeCoT is applied to RGB+A(GT). The prompt explicitly asks the model to interpret the external gaze cue before choosing an answer, yet the model can still misread a correct arrow.

Example 1 (ground truth: ***remote control****).* The model returns **coffee**, stating that the arrow tip lands "directly on the clear plastic cup containing a dark liquid," although the ground-truth arrow points to the **remote control**.

Example 2 (ground truth: ***remote control****).* The model returns **water**, stating that the arrow "points directly at the water bottle on the table," although the ground-truth arrow points to the **remote control**.

These cases show that structured gaze reasoning can produce a target description that contradicts the overlay even when the gaze cue itself is correct.

9 Implementation Details

Depth Anything V2 [5]. ViT-Large (*vitl*) encoder, input size 518. Per-image depth is clipped to the 1st and 99th percentiles of valid positive values. Visual depth plates and depth-colored arrows map this range to 8-bit values and use OpenCV's JET pseudocolor colormap; textual depth coordinates scale the same range to $[0, 1]$.

UniGaze [2]. Input resized to 448×448. We take the gaze direction from the projected 3D face center and find its first intersection with scene geometry in the depth map (first occluder; fallback: image boundary); see Section 3.

YOLOv10L [4]. Confidence threshold 0.25, NMS IoU 0.45 (Ultralytics defaults). Heuristic baseline: exclude "dining table" on all tracks and "person" on FG-GRI only (never a FG-GRI ground-truth answer).

Heuristic baseline scoring. The heuristic baseline selects the candidate whose YOLOv10L bounding-box center is nearest to the predicted endpoint. We score it by manual review—one author inspects each selected box and marks the instance correct if the box covers the ground-truth gaze target. Because the correctness criterion (box coverage) differs from the LMM protocol (one named candidate, exact-match parsing), this accuracy is a rough reference point for how far a purely geometric pipeline gets, not a strictly comparable number.

LMM decoding. Proprietary models were queried via commercial APIs in April–May 2026. GPT calls use `gpt-4.1`, `gpt-5.1`, and `gpt-5.2`. Gemini calls use `gemini-2.5-flash` and `gemini-3.1-pro-preview`.

Decoding uses temperature 0 wherever the provider accepts it, falling back to the provider default only for models that reject the parameter; each instance is queried once per condition. Answers are recovered by a JSON-constrained response or a post-hoc parse of the `answer` field, depending on provider support. Open-source models use the same prompt templates and parser. The open-source checkpoints are Qwen3-VL-8B-Instruct, Qwen3.5-27B, Llama-3.2-11B-Vision-Instruct, and Gemma-4-26B-a4b-it, matching the names in the main paper.

Data and code availability. The GRI-Bench release is annotation-only: FG-GRI and CG-GRI candidate lists, target labels, observer boxes, ground-truth gaze-arrow coordinates, evaluation splits, prompt templates, and scripts. Predicted cues such as UniGaze rays are experimental conditions and are not redistributed. No source-dataset pixels are included. Frames are recovered from Zhang et al. Stimuli, VideoAttentionTarget, and GazeFollow via the released `source_relpath` join key. The construction script copies those frames locally and stamps the unfilled red observer box on every CG-GRI image; a companion script overlays the released ground-truth arrows.

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